AIM:- Aim :- Using Packet Tracer to create a basic network: one server, two PC computers and one smartphone, one laptop using appropriate network wire and checking proper connections.

Step 1: Set up Packet Tracer

- Open **Cisco Packet Tracer**.
- From the **Devices** section (bottom left), add the following:
- **3 Routers**: Drag and drop 3 routers onto the workspace. You can use **Router 1**, **Router 2**, and **Router 3**.

- **3 Switches**: Each router needs a switch to connect the PCs. Add 3 switches.

- **9 PCs**: You need at least 3 PCs for each router's network. Add 9 PCs in total.

STEP 2:-Connect the Devices

Inter-Router Connections:

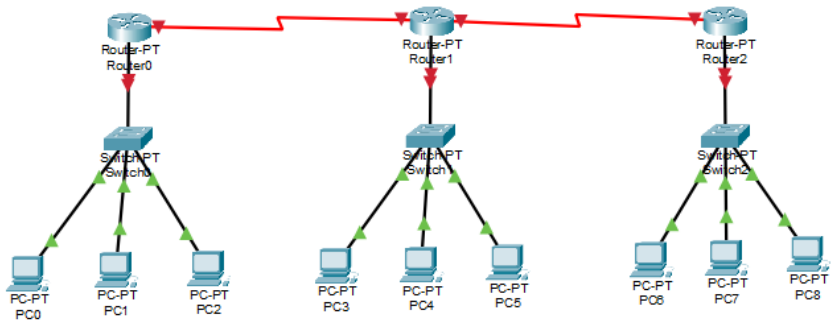
- Use the Copper Straight-Through Cable to connect the routers:
    - Connect Router 1 to Router 2.
    - Connect Router 2 to Router 3.

Router to Switch Connections:

- Connect each router to a switch using a Copper Straight-Through Cable.
    - Router 1 connects to Switch 1.
    - Router 2 connects to Switch 2.
    - Router 3 connects to Switch 3.

Switch to PC Connections:

- Connect each switch to 3 PCs using **Copper Straight-Through Cable**.
    - **Switch 1** to 3 PCs.
    - **Switch 2** to 3 PCs.
    - **Switch 3** to 3 PCs.

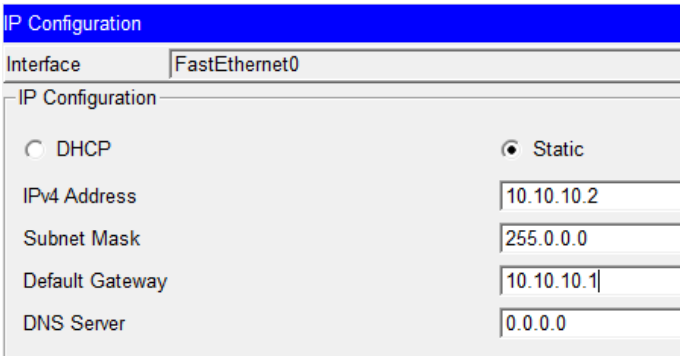


Step 3: Assign IP Addresses

3.1. Configure IP on PC0.

1. Click PC0
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 10.10.10.2
    - Subnet mask : 255.255.255.0
    - Default Gateway : 10.10.10.1

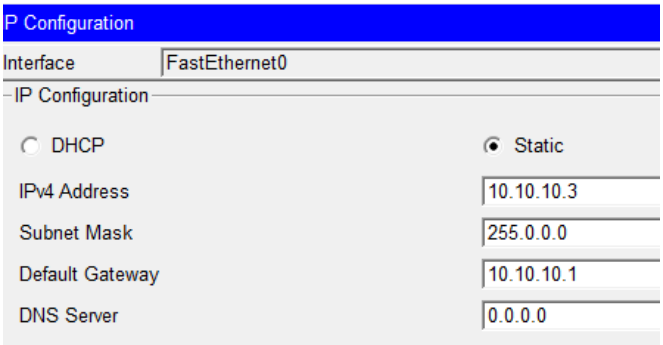
OUTPUT:-



3.2. Configure IP on PC1.

1. Click PC1
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 10.10.10.3
    - Subnet mask : 255.255.255.0
    - Default Gateway : 10.10.10.1

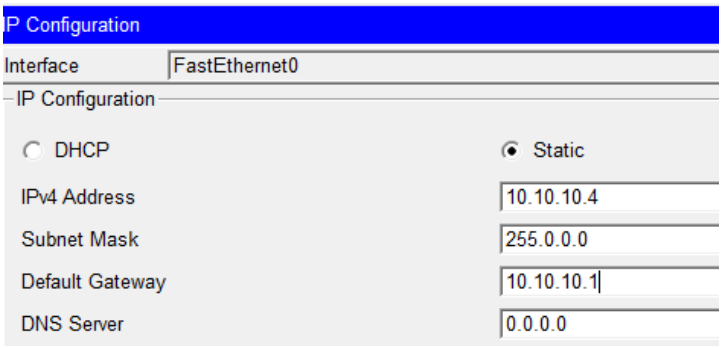
Output:-



3.3. Configure IP on PC2.

1. Click PC2
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 10.10.10.4
    - Subnet mask : 255.255.255.0
    - Default Gateway : 10.10.10.1

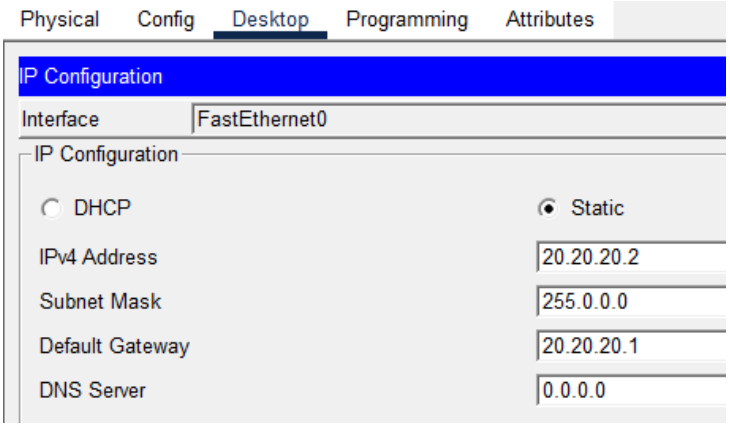
Output: -



3.4. Configure IP on PC3.

1. Click PC3
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 20.20.20.2
    - Subnet mask : 255.255.255.0
    - Default Gateway : 20.20.20.1

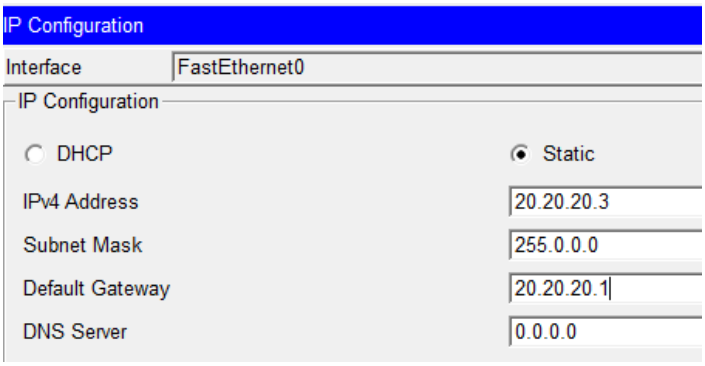
Output:-



3.5. Configure IP on PC4.

1. Click PC4
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 20.20.20.3
    - Subnet mask : 255.255.255.0
    - Default Gateway : 20.20.20.1

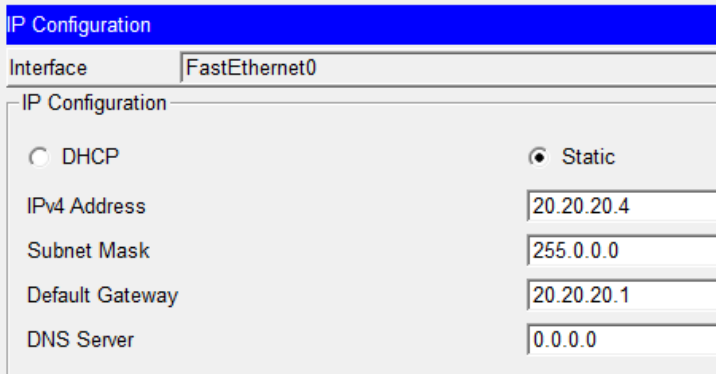
Output:-



3.6. Configure IP on PC5.

1. Click PC5
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 20.20.20.4
    - Subnet mask : 255.255.255.0
    - Default Gateway : 20.20.20.1

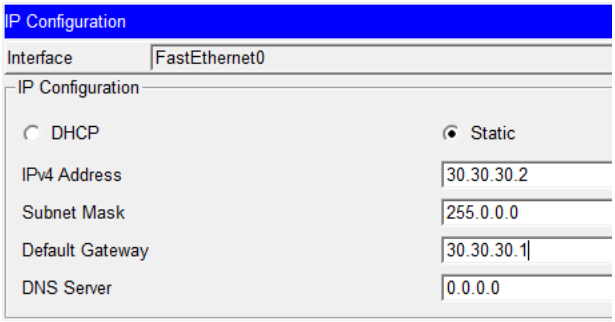
Output:-



3.7. Configure IP on PC6.

1. Click PC6
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 30.30.30.2
    - Subnet mask : 255.255.255.0
    - Default Gateway : 30.30.30.1

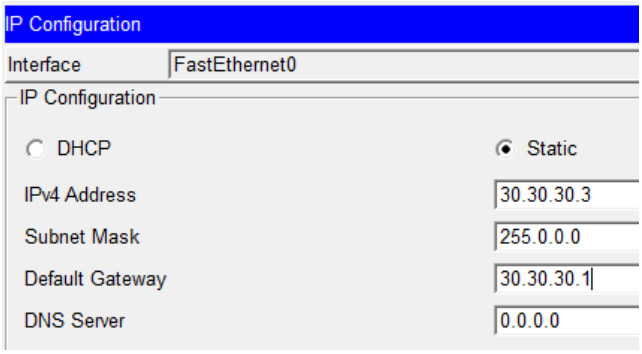
Output:-



3.8. Configure IP on PC7.

1. Click PC7
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 30.30.30.3
    - Subnet mask : 255.255.255.0
    - Default Gateway : 30.30.30.1

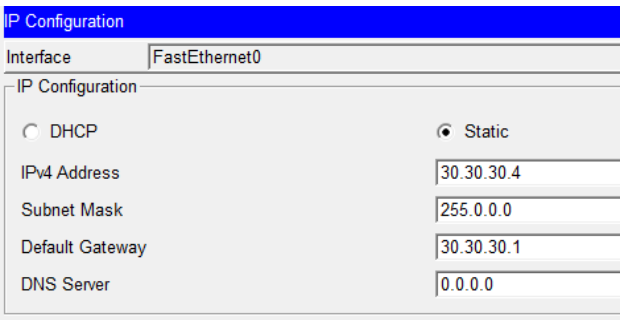
Output:-



3.9. Configure IP on PC8.

1. Click PC8
2. Go to Desktop -> IP configuration
3. Set:
    - IP address : 30.30.30.3
    - Subnet mask : 255.255.255.0
    - Default Gateway : 30.30.30.1

Output:-

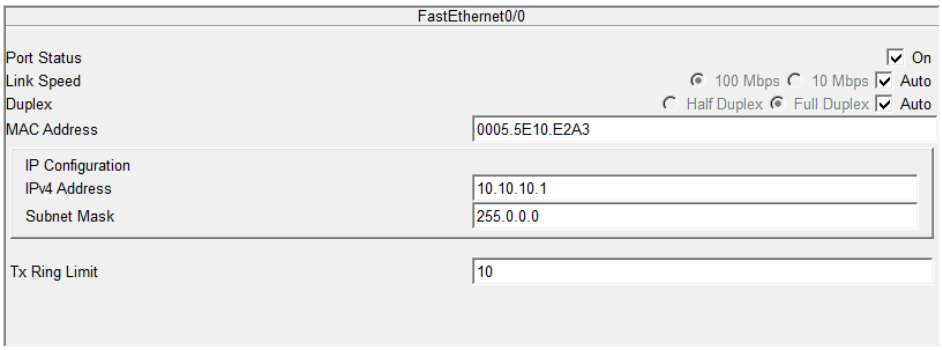


Step 4 :- Configure IP address on Routers.

4.1.1. Configure IP Address on Router0

1. Click Router0.
2. Go to config -> FastEthernet0/0.
3. Turn On the Port Status.
4. Set:
    - IP address : 10.10.10.1
    - Subnet mask : 255.0.0.0

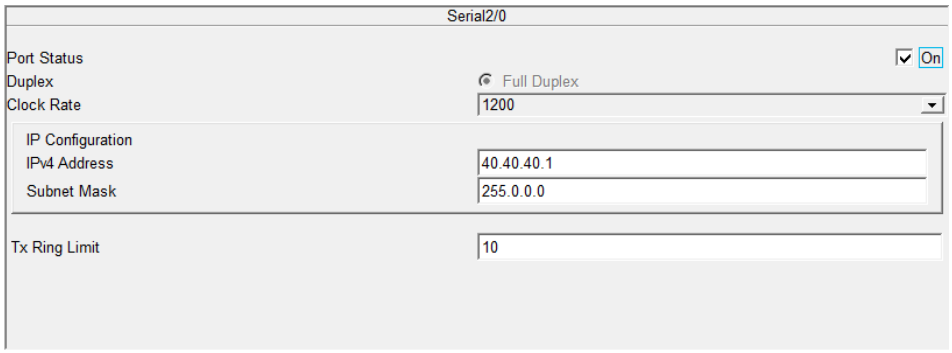
Output:-



4.1.2. Configure Serial Interface on Router0

1. In Router0 , select the Serial2/0 interface.
2. Turn On the Port Status.
3. Set the following values:
    - IP Address : 40.40.40.1
    - Subnet Mask : 255.0.0.0

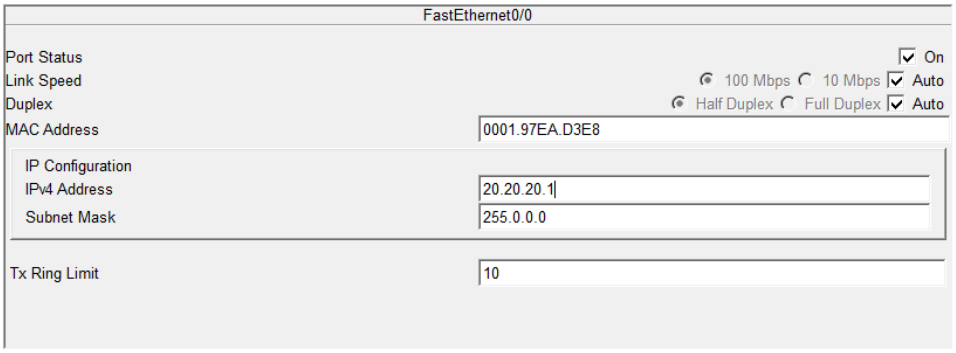
Output:-



4.2.1. Configure IP Address on Router1

1. Click Router1.
2. Go to config -> FastEthernet0/0.
3. Turn On the Port Status.
4. Set:
    - IP address : 20.20.20.1
    - Subnet mask : 255.0.0.0

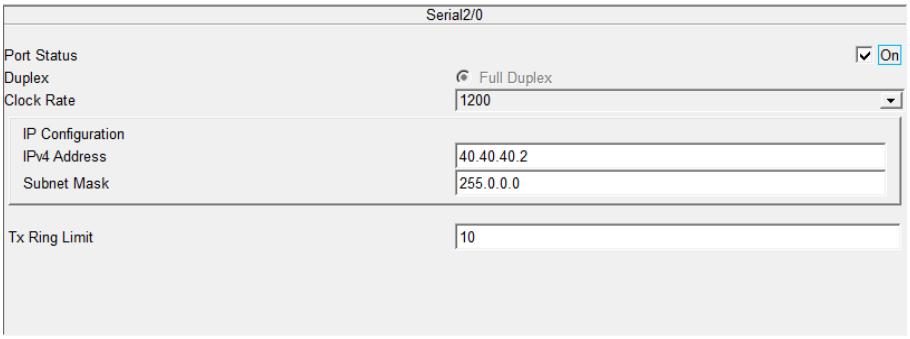
Output:-



4.2.2. Configure Serial Interface on Router1 (Serial2/0)

1. Click Router1.
2. Go to Config -> Serial2/0.
3. Turn On the Port Status.
4. Set:
   - IP Address : 40.40.40.2
   - Subnet Mask : 255.0.0.0

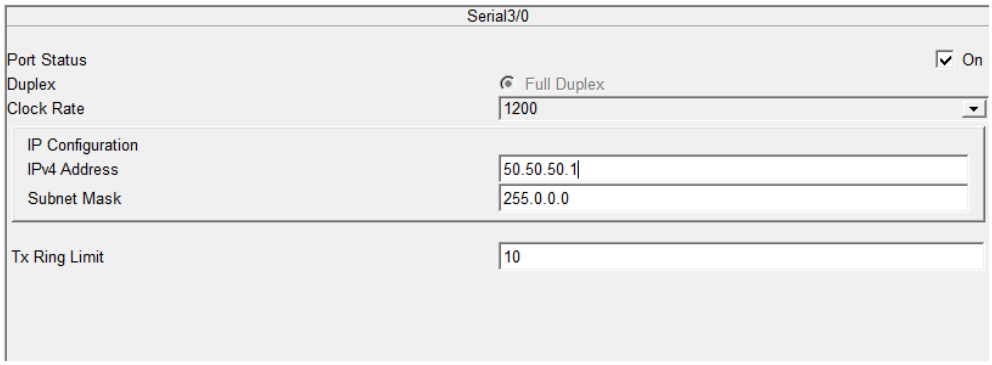
Output:-



4.2.3. Configure Serial Interface on Router1 (Serial3/0)

1. Click Router1.
2. Go to Config → Serial3/0.
3. Turn On the Port Status.
4. Set :
   - IP Address : 50.50.50.1
   - Subnet Mask : 255.0.0.0

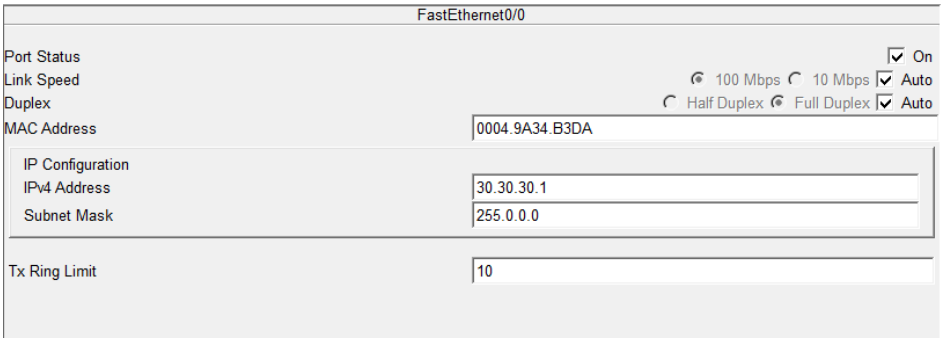
Output;-



4.3.1. Configure IP Address on Router2

1. Click Router2.
2. Go to Config -> FastEthernet0/0.
3. Turn On the Port Status.
4. Set:
    - IP Address : 30.30.30.1
    - Subnet Mask : 255.0.0.0

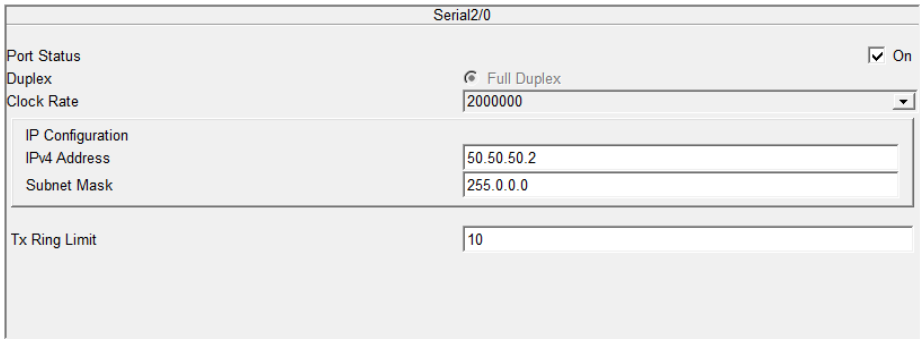
Output:-



4.1.7. Configure Serial Interface on Router2

1. Click Router2.
2. Go to Config -> Serial3/0.
3. Turn On the Port Status.
4. Set the following values:
    - IP Address : 50.50.50.2
    - Subnet Mask : 255.0.0.0

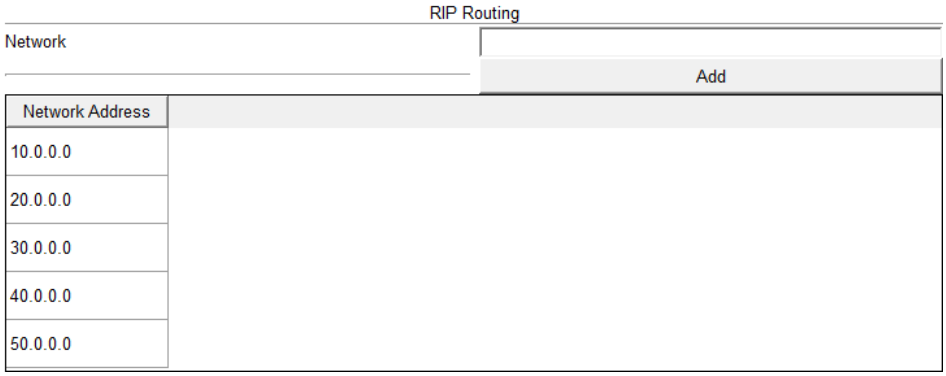
Output:-



Step 5 :- Enable RIPv1 on Router0

1. Click Router0.
2. Go to Config -> RIP.
3. Add the following network address:
    - 10.0.0.0
    - 20.0.0.0
    - 30.0.0.0
    - 40.0.0.0
    - 50.0.0.0
4. Do the same for Router1 and Router2.

Output:-



Step 6 :- Test Connectivity with PDU.

1. Select Add Simple PDU (envelope icon) from the toolbar.
2. Click on PC0 (Source Device).
3. Click on PC4 (Destination Device).
4. Check that the final status is Successful.

Output:-

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC4	ICMP		0.000	N	0	(edit)

Do the same for other PCs also.

Result:-

- RIPv1 routing is successfully configured and communication between all networks is verified.